

ASSIGNMENT On
Computer Organization And
Design.

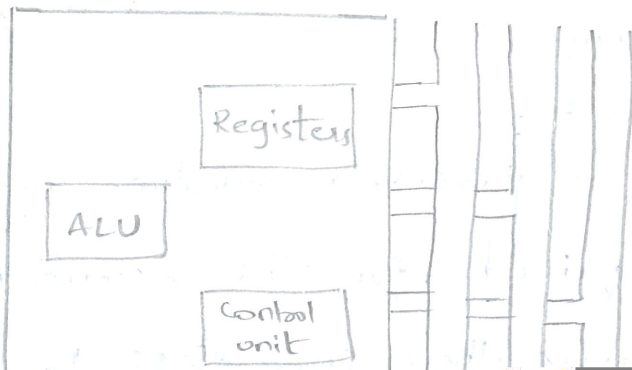
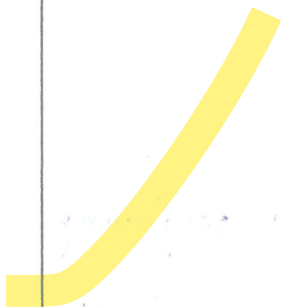
UCSH-301

Processor Organization

Processor requirements:

The organisation of a processor revolves around performing essential tasks required for the execution of instructions.

The processor follows these steps to execute an instruction:



Register Organization:

- Computer system employs a memory hierarchy. At higher levels of the hierarchy, memory is faster, smaller, and more expensive (volatile).

This figure is a slightly more detailed view of the processor. The data transfer and logic control paths

are indicated by solid and dashed lines respectively.

- General purpose: It can be assigned to a variety

• **Condition Codes:** It also referred to as flags. Bits set by the processor hardware as the result of operations.

Advantages	Disadvantages

Control & status registers:

There are a variety of processor registers that are employed to control the operation of the processor.

AC															
1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
1	1	1	0	0	0	1	0	1	X	1	X	X	1	1	.

- Many processor designs include a register or set of registers, often known as the program status word (PSW), that contain status information. It contains

• Instruction Cycle:

Includes the following stages:

Fetch

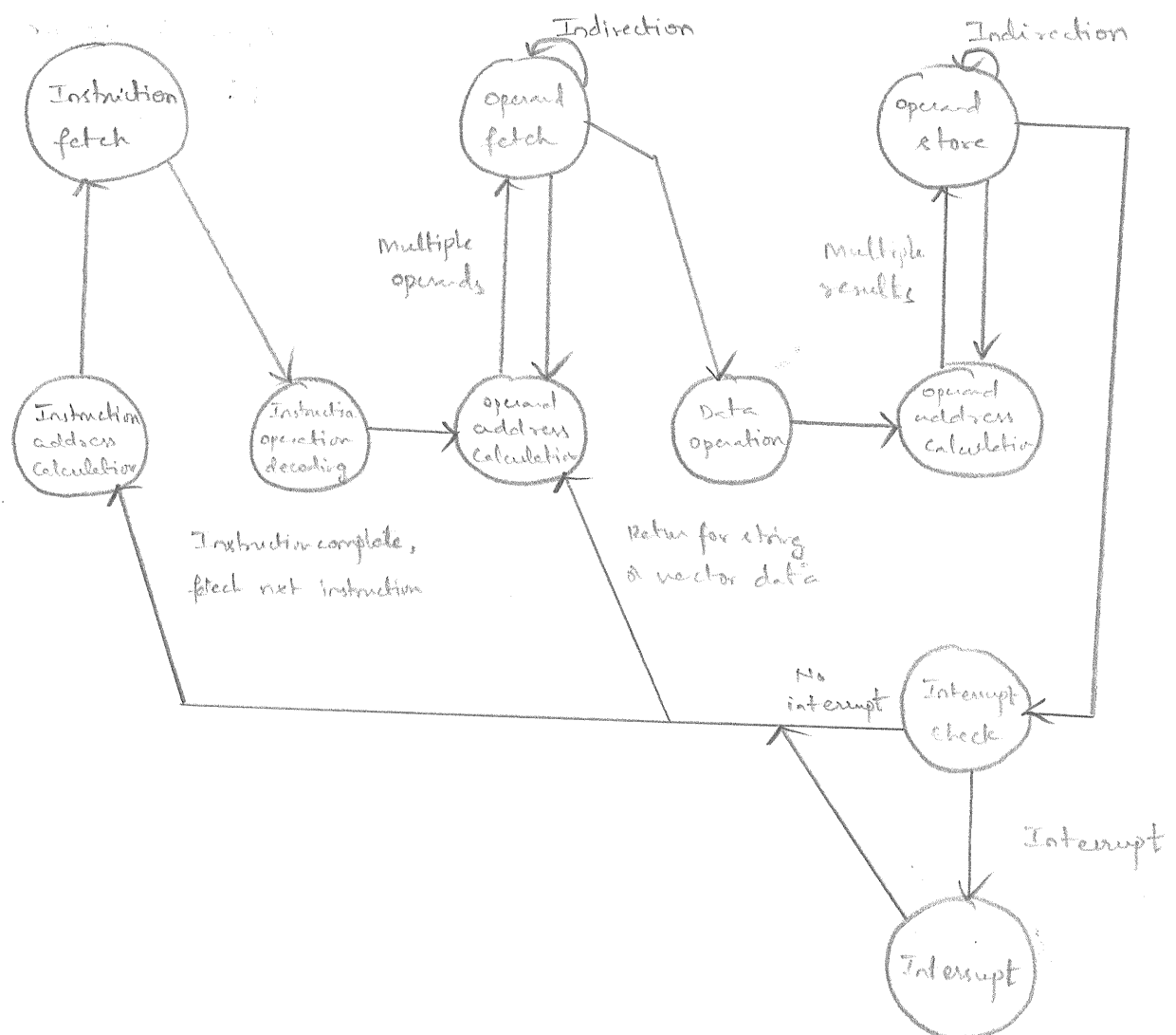
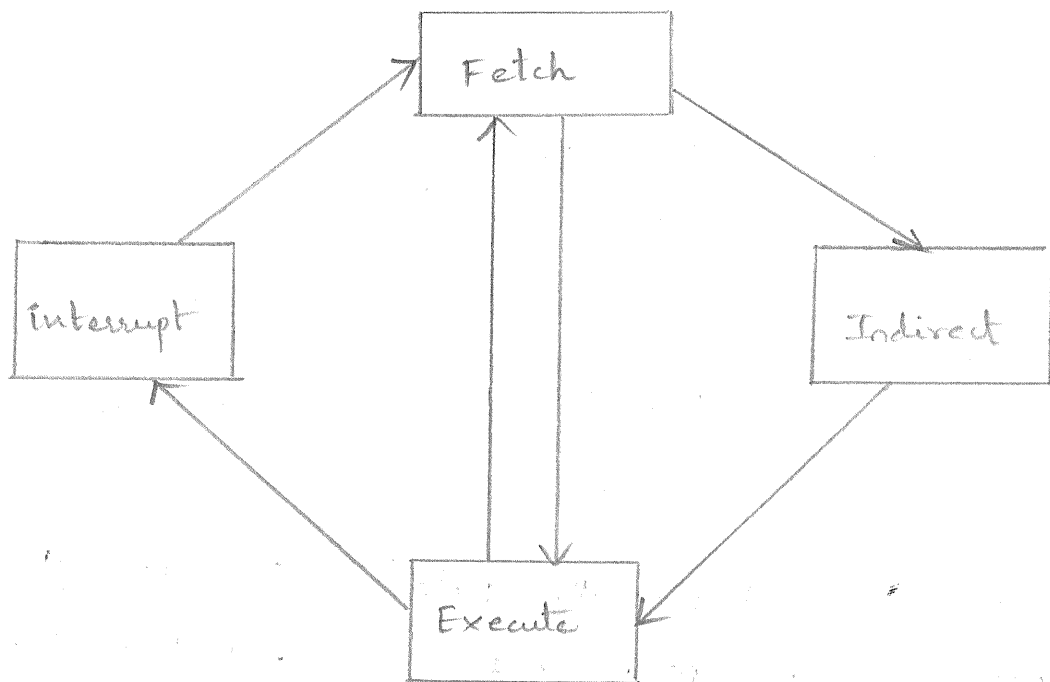
Execute

Interrupt

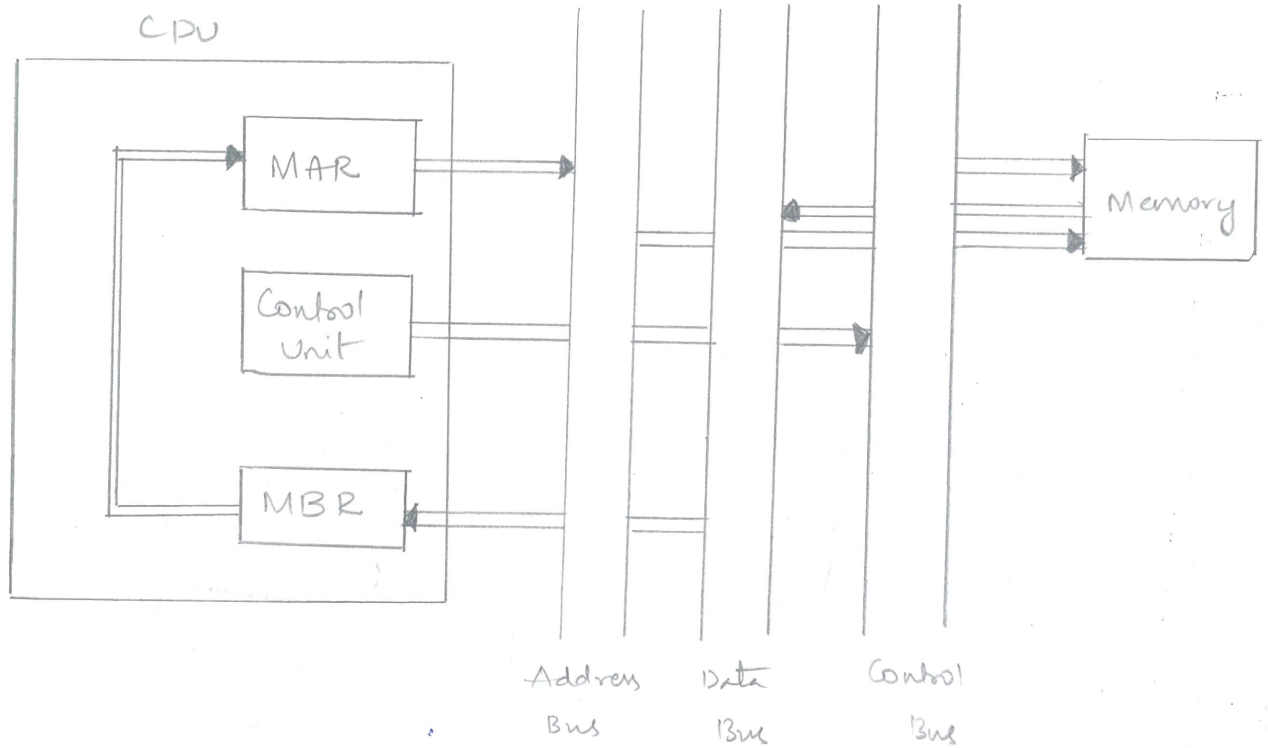
Read the next

Interpret the opcode

If interrupts are



• Data flow, Indirect cycle.



Address Bus Data Bus Control Bus

Examples of Microprocessor

→ It asserts the RD signal to read the instruction (MVA)

.. Internal Memory : In the 8085 microprocessor, internal memory can refer to general purpose registers (B, C, D, E, H, L).

→ left shift: If $A = 05H$ (Binary: 00000101), left shifting by 1 bit results in $A = 0AH$ (Binary: 00001010) which is equivalent to multiplying 5 by 2.

→ Right shift: It could divide the value by 2.

